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- An Empirical Analysis of the Performance of Sustainable Investing

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Is Sustainability to the World Sustainable to Your Wealth?

- An Empirical Analysis of the Performance of Sustainable Investing

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Abstract

This study empirically examines the return and risk of companies with different ESG scores and what level of ESG engagement yields the most efficient return-to-risk. We run a Fama-French 5-Factor regression on five different portfolios with different ESG engagement using monthly data for the companies in the S&P500 over the last 20 years. We find that companies with lower ESG scores in general generate lower returns with the environmental pillar as the driving factor. We then extend our analysis by conducting a fixed effect regression to find the relationship between ESG and risk. We find that an increase in ESG engagement yields lower volatility. Similarly, the environmental pillar is the driving factor of this effect but interestingly we find the social pillar to increase both volatility and idiosyncratic risk. We then compare the return-to-risk ratio of our five portfolios and find that the increase in returns for companies with low ESG scores more than compensates for the higher risk. Therefore, investing sustainably provides less risk for the investors, but is less efficient due to the much lower return.

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1 Introduction

ESG investing is playing an increasingly important role in financial investment decisions for both private- and institutional investors. Ethical considerations are a high motivational factor for investing in social responsible companies as more private investors care about factors such as the climate, human rights and being sustainable for future generations (IISD, 2020). Furthermore, the increasing interest in ESG investing also reflects beliefs that environmental, social, and corporate governance issues can have a significant impact on the long-term performance of companies, which is mostly considered among institutional investors (Boffo and R. Patalano, 2020). However, for most investors the primary goal of investing is to receive a return and to grow their fortune. This can either be to increase their overall wealth or just to combat the decreasing purchasing power from inflation. Given that several empirical studies have suggested that companies highly engaged in ESG activities tend to underperform in terms of returns (Bannier et al., 2019; Fabozzi et al., 2008; de Haan et al., 2012), investors need to consider whether the higher return of companies with lower ESG ratings compensates for compromising their ethical considerations.

The primary objective of this paper is to examine whether investing in companies with high ESG scores yield a more efficient portfolio. Thereby we reexamine the impact of socially responsible investments on stock performance to determine the potential advantages of investing in highly rated ESG companies. Furthermore, we seek to determine whether the three pillars of the ESG score affect stock performance equally or if some measures have a larger impact. We extract monthly data on the 500 companies in the S&P500 index from Bloomberg and Refinitiv Eikon from January 2003 to January 2023.

We first investigate the relationship between ESG scores and returns by estimating a Fama-French 5-Factor model. Our findings are consistent with previous studies in the literature as we find that companies with high ESG ratings tend to generate lower returns. Additionally, we run the same regression to find the relationship between return and environmental-, social-, and governance scores, respectively. We find that an increase in environmental score leads to the largest decrease in return and therefore is the driving factor in the negative relationship between ESG scores and return.

Due to return being a pay-off for higher risk and multiple studies highlighting the advantages of lower risk of socially responsible investments, we further examine the relationship between ESG scores and equity risk. We run a fixed effect estimation on different risk measures and find that higher ESG scores decrease risk. However, our findings do not support a consistent negative relationship between all the pillars in ESG when we separate them in our estimation. Specifically, we observe a strongly negative and statistically significant relationship between the environmental score and a firm's total risk, a smaller yet still negative effect from the governance score, but surprisingly, a highly positive estimate for the social factor. This is in accordance to other literature for the U.S. such as (Bannier et al., 2019), however, in contradiction to literature for Europe (Sassen et al., 2016) that finds a negative effect from the social pillar. Geographical factors should therefore be taken into consideration when interpreting the results.

Lastly, we compare the benefit of higher return to the higher associated risk of investing in companies with lower ESG ratings. We find the low ESG portfolio to be the most efficient due to the highest return-to-risk ratio as the high return more than compensates for the increased risk.

The remainder of the paper is structured as follows: In Section 2 we first introduce the different ESG measures and the general portfolio theory used throughout the paper. We then go through the relevant theory behind our regression models that includes the Fama-French 5-Factor model and fixed effect model. In Section 3 we go through the data used in the paper including descriptive statistics. Furthermore, we go into detail about how we construct our different portfolios. In Section 4 we first run the Fama-French 5-Factor regression on our data to find the relation between ESG scores and return. We then extend our analysis and run a fixed effect model on our data to examine the relation between ESG scores and different risk measures. Lastly, we compare the results of our two regressions. In Section 5 we discuss our method, results and possible implications. Finally, in Section 6 we conclude our paper.

2 Theory

2.1 ESG Investments

ESG investments are investments in companies or assets that prioritise environmental, social, and governance (ESG) factors in their decision-making processes. In recent years, there has been a significant increase in the popularity of ESG investing. By considering ESG factors, investors can gain insights into the potential risks and opportunities associated with a company's environmental practices, social impact, and corporate governance structure. Companies that demonstrate sustainable business practices and are well-positioned might be able to deliver superior long-term returns (Boffo and R. Patalano, [2020](#)).

Refinitiv Eikon is a financial data platform that provides a wide range of data and tools for people practicing finance, including ESG data. Refinitiv Eikon's ESG data covers a broad range of environmental, social, and governance metrics and scores. The scores can help investors identify which companies that have a strong performance on the three factors and which companies may be lagging in their commitments to sustainability.¹

The environmental (E) pillar refers to the usage of energy, water, and resources by firms, as well as their management of these resources, and the discharge of carbon emissions and waste. This component evaluates how the actions of firms impact the environment and contribute to climate change and how well they innovate to improve these actions in the future (Refinitiv, [2022](#)).

The social (S) pillar evaluates a company's interactions with its stakeholders and their reputation, particularly with the people and community where the business operates. This includes employees, customers, subcontractors, and government entities. The social component includes labor relations, employee satisfaction, diversity and inclusion, the firm's adherence to fundamental human rights, and safeguarding the health and safety of its employees. Additionally, it encompasses product responsibility, including product safety, consumer protection, and data privacy (Refinitiv, [2022](#)).

¹An overview of the structure of the scores from Refinitiv Eikon can be found in Appendix [A1](#) & [A2](#).

The governance (G) pillar of a business pertains to the internal processes, controls, reporting, and practices that the firm implements to govern itself, satisfy stakeholders' needs, comply with laws, and have an efficient decision-making process. This component also evaluates the firm's transparency in reporting to the public and their overall CSR (Corporate Social Responsibility) strategy. It addresses the diversity, compensation, and independence of the board and shareholders (Refinitiv, 2022).

Using these three pillar scores we are able to compute the well-known ESG score that is the relative sum of the pillar weights. These weights vary across industries for both the environmental pillar and the social pillar but remains the same for the governance pillar. The pillar weights are normalised to percentages between 0 and 100, thus, making the ESG score between 0 and 100 as well, where 100 is the best (Refinitiv, 2022). However, as mentioned the weights of the environmental- and social pillar weights differentiate across industries, thus, making the overall ESG score different across industries as well. For all the numeric data points in the environmental and social weights, a comparison is made to the relative median value specific to the industry. This comparison helps determine whether a company has performed better or worse than its peers within the same industry. Because of this construction, it is important to be cautious when comparing ESG scores across industries. For instance, a company could score well in an industry with generally poor performance, yet still have higher CO_2 emissions compared to a company with a lower ESG score in an industry that performs well across various ESG measures.

Furthermore, Refinitiv Eikon also reports a Combined ESG score or ESGC score that incorporates a controversies score. The controversies score is calculated on the base of 23 ESG controversy topics and goes from 0 to 100 where 100 is the best. If a company is involved in a scandal they receive a lower controversies score which decreases their ESGC score. It should be noted that the controversies score takes into account the market cap bias where large-cap companies receive more media attention compared to small cap companies (Refinitiv, 2022).

2.2 Portfolio Theory

Our analysis of sustainable investments will be based on modern portfolio theory. The theory will give insights on how to find and measure return and risk. For this, we will begin by introducing the widely used model in finance, namely the Capital Asset Pricing Model

(CAPM). This provides a simple way to estimate the expected return on an investment based on its risk. Furthermore, we will go through different risk measures in order to get a better understanding of the different natures of a firm's risk.

2.2.1 Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is a financial model used to determine the expected return on an investment, based on the risk associated with that investment. The model assumes investors are risk-averse, so they select efficient portfolios that maximize an investment's expected return given a level of risk. When an investment has a higher level of risk, investors demand a higher expected return known as the risk premium. Thus, CAPM demonstrates the trade-off between risk and return

The CAPM suggests that the required expected return is:

$$E[R_{i,t}] = r_f + \beta_i(E[R_M] - r_f) + \varepsilon_{i,t} \quad (2.1)$$

meaning the expected return equals the risk free rate plus β times the market risk premium. In perfect markets the idiosyncratic error term $\varepsilon_{i,t}$ is expected to be 0 (Berk and DeMarzo, 2020).

β in CAPM is a measure of an asset's volatility in relation to the overall market. It is defined as the covariance between the portfolio's return and the market return divided by the variance of the market return, $\beta_i = \frac{\text{cov}(r_i, r_m)}{\text{var}(r_m)}$. It measures the sensitivity of an asset's return in relation to the market's return. A $\beta = 1$ indicates that the asset has the same level of volatility as the overall market. A $\beta > 1$ indicates that the asset is more volatile than the market, while a $\beta < 1$ indicates that the asset is less volatile than the market. The CAPM, thus, describes the relationship between the expected return for an asset and the associated systematic risk, β . This indicates that the β is the only relevant factor in determining an asset's price, according to CAPM, because investors are exclusively rewarded (in the form of higher expected returns) for the systematic risk. Systematic risk is the risk of the market, which you cannot get rid of by diversifying your portfolio. Due to the assumption that investors are risk averse, they will get compensation for the systematic risk.

The CAPM is often shown graphically to illustrate and evaluate the trade-off between systematic risk and return for an asset. This is called the security market line (SML), which

can be found in Appendix A3. It is represented as a straight line where the expected return is on the y-axis and β on the x-axis, where the slope of the line is determined by the market risk premium. In theory, it should be a straight line because according to CAPM, β is the only thing that determines the excess return of an individual asset. However this is found not be true in all cases. The expected return of an assets can be different from what is predicted from β due to shocks. This deviation is incorporated into the CAPM as the intercept α :

$$\alpha_i = E[R_{i,t}] - r_f - \beta_i(E[r_m] - r_f)$$

As α is the deviation from the predicted return of SML it is often used as a measure of an asset's performance relative to the market. Thus, if α is positive, it means that the asset has yielded a larger excess return than predicted by the SML. Assets that lie above the SML are considered to be undervalued, since they offer a higher expected return than what would be expected given their level of risk. Assets that lie below the SML are considered overvalued, since they offer a lower expected return than what would be expected given their level of risk (Berk and DeMarzo, 2020).

2.2.2 Risk Measures

Greater risk implies uncertainty regarding future outcomes, which can significantly impact the value of a firm. This is primarily due to the potential increase in the cost of capital, uncertainty surrounding future cash flows, and challenges in making capital budgeting decisions, among other factors (Luo and Bhattacharya, 2009). There are several risk measures used in finance to quantify the potential loss of an investment. In order to quantify the relationship between risk and return, we need to think carefully about how to measure risk. There are multiple types of risk measures that we can apply to examine the risk of sustainable investments.

In our analysis, we mainly focus on two risk measures: stock price volatility and idiosyncratic risk. The stock volatility is the firm's total risk and is often calculated as the standard deviation using a stock's daily returns:

$$\sqrt{Var(R_{i,t})} = \sqrt{\frac{1}{T-1} \sum_{i=1}^T (R_{i,t} - \bar{R}_i)^2}$$

where $R_{i,t}$ is the daily return of stock i at time t , $\bar{R}_i$ is the mean of the returns of stock i and T is the number of observations of daily stock returns through the year. It measures how

much the asset's value deviates from its mean on average. A higher standard deviation will result in higher risk. The stock volatility contains the two risk components: systematic and idiosyncratic risk, where we will look further into the idiosyncratic risk.

The idiosyncratic risk, or the unsystematic risk, is purely firm-specific, as it is unrelated to the market. This type of risk relates to good or bad news about the firm itself. Thus, it is possible to diversify away by investing in a broad range of different firms, which decreases the effect of firm-specific news on the overall portfolio. As it is possible to diversify this risk away, it is assumed that it does not affect the return of stocks, per CAPM. However, literature such as (Goyal et al., 2002; Lee and Faff, 2009) find a positive relation between idiosyncratic risk and return and it is therefore still relevant to include this risk measure. As the idiosyncratic risk is only firm-specific it can be calculated as the standard deviation of our idiosyncratic error term in CAPM (Equation 2.1): $\sigma_{\varepsilon_{i,t}} = \sqrt{\text{Var}(\varepsilon_{i,t})}$. By taking the variance of both sides of CAPM it yields:

$$\begin{aligned} \text{Var}(R_{i,t}) &= \text{Var}(r_f + \beta_i(R_M - r_f) + \varepsilon_{i,t}) \\ \sigma_{R_{i,t}}^2 &= \underbrace{\beta_i^2 \sigma_{R_M}^2}_{\text{Systematic Risk}} + \underbrace{\sigma_{\varepsilon_{i,t}}^2}_{\text{Idiosyncratic Risk}} \\ \sigma_{\varepsilon_{i,t}} &= \sqrt{\sigma_{R_{i,t}}^2 - \beta_i^2 \sigma_{R_M}^2} \end{aligned}$$

We are therefore able to see that the idiosyncratic risk comes down to the excess volatility of a stock compared to the market. This also shows intuitively how a diversified portfolio consisting of many stocks that have a volatility closer to the market will yield a lower idiosyncratic risk.

2.3 Fama-French 5-Factor Model

CAPM is often found not to precisely predict the expected return of an asset. If there are systematic deviations from the Security Market Line (SML), the market portfolio cannot be the efficient portfolio. Thus, some risk compensation may not be fully captured by the market risk and therefore, the model should be extended. We therefore further extend the model to the Fama-French 5-Factor model in order to take into account additional factors beyond the market risk premium.

The model extends the CAPM attempting to explain average stock returns by taking into account additional factors beyond the market risk premium. Thereby it is no longer only based on the historical changes in stock prices. The regression model is a standard OLS and looks as follows:

$$R_{i,t} - r_{ft} = \alpha_i + \beta_{1,i}(R_{Mt} - r_{ft}) + \beta_{2,i}SMB_t + \beta_{3,i}HML_t + \beta_{4,i}RMW + \beta_{5,i}CMA + \varepsilon_{i,t} \quad (2.2)$$

where R_{it} represents the total return of a portfolio i at time t , and R_{ft} is the risk free rate at time t , while R_{Mt} is the total return of the market portfolio at time t .

SMB (Small Minus Big) indicates the average return of three small company portfolios minus the average return of three big company portfolio.

$$SMB = \frac{1}{3}(\text{Small Value} + \text{Small Neutral} + \text{Small Growth}) - \frac{1}{3}(\text{Big Value} + \text{Big Neutral} + \text{Big Growth})$$

This is the size factor and it is based on the idea that small-cap stocks tend to outperform large-cap stocks in return over time, as they have more room for growth and are more risky.

HML (High Minus Low) is the average return of two value portfolios minus the average return on two growth portfolios.

$$HML = \frac{1}{2}(\text{Small Value} + \text{Big Value}) - \frac{1}{2}(\text{Small Growth} + \text{Big Growth})$$

The HML exploits the fact that historically firms with higher book valuation have outperformed firms with lower book valuation relative to the market valuation. It is based on the idea that stocks that are undervalued by the market (low price-to-book ratio), tend to outperform growth stocks with high price-to-book ratios, which are overvalued by the market (French, 2023).

RMW (Robust Minus Weak) is the difference between the returns of diversified portfolios of stocks with a robust and weak profitability, respectively (Fama and French, 2014).

$$RMW = \frac{1}{2}(\text{Small Robust} + \text{Big Robust}) - \frac{1}{2}(\text{Small Weak} + \text{Big Weak})$$

Theoretically, stocks with high expected profitability should yield a higher average return than stocks with low expected profitability (Hou et al., 2015). Thus, the RMW-factor takes

into account the difference there might be in average returns between companies with high and low profitability where a positive estimate signifies a greater return from companies with a higher profitability.

CMA (Conservative Minus Aggressive) is the difference between the returns of stocks of firms that engage in low and high investments, categorised as conservative and aggressive, respectively (Fama and French, 2014).

$$CMA = \frac{1}{2} (\text{Small Conservative} + \text{Big Conservative}) - \frac{1}{2} (\text{Small Aggressive} + \text{Big Aggressive})$$

Theoretically, high investment stocks should yield a lower expected average return than stocks with low investments (Hou et al., 2015). Thus, the CMA-factor takes into account the difference in average returns between companies with low and high investments where a positive estimate signifies a greater return from more conservative companies with smaller investments.

Lastly, the α in the regression is similar to the α mentioned in Section 2.2.1 and is also known as Jensen's α and is our primary variable of interest. Jensen's α describes the abnormal return of the specific stock or portfolio and is often used as a measure of how well a portfolio manager performs compared to the market. Thus, by comparing the α 's of different portfolios one is able to assess how well all the portfolios perform compared to the market. Furthermore, the α can also be seen as the remainder of the return that cannot be explained by the rest of the variables. By using this extended regression of the CAPM it provides us with a better explanation of the average returns as it yields a numerically smaller intercept α (Fama and French, 2014). Thus, most of the excess return will be explained by the factors added to the model ensuring that we are able to use sustainability performance to describe the remainder of the return (Bannier et al., 2019). By constructing different portfolios based on ESG scores we are then able to compare the α 's of these portfolios where the difference between α 's then theoretically only consists of the difference in ESG scores. We can therefore determine whether portfolios consisting of firms with higher ESG scores yield higher abnormal returns compared to their lower performing counterparts. Thus, by comparing the α 's of different ESG-portfolios we are able to estimate the effect ESG scores have on the return of a portfolio.

2.4 Fixed Effect Model

The Fama-French 5-Factor model gives a good insight into the relationship between ESG and return, but lacks to incorporate estimates of the underlying risk of the abnormal return. We intend to dig deeper into the connection between ESG and various risk factors to determine if sustainable investments could be associated with safer investments. To achieve this, we employ a fixed effect estimation, which allows us to conduct a more extensive analysis of the relationship between ESG and different risk factors.

Panel data consists of observations on a set of individuals observed over time. It is the same individuals (firms) we observe over time, allowing us to study changes in variables over time as well as individual differences in those changes. Some of the most common estimation methods to use when dealing with panel data are the fixed effect and the random effect. In panel data, there may be unobservable factors or characteristics specific to each firm that can affect both the dependent variable and the explanatory variables, called individual-specific effects (Wooldridge, 2001). We will now go more into depth with how we deal with panel data and some of the potential biases that may arise.

We extend our analysis by analysing the relationship between ESG scores and volatility and idiosyncratic risk, respectively. Since we are dealing with panel data we need to account for unobserved heterogeneity or time-invariant characteristics of each firm. This is to control for the fact that we believe there might be unobserved differences in the firms that are expected to affect the volatility and idiosyncratic risk that cannot be directly observed or measured. This is due to the fact that the same firms are observed at multiple points in time, and there may be factors that influence our outcome variable that are constant over time and unique to each firm (Wooldridge, 2018). Thus, using a fixed effect regression we are able to control for any stable source of difference between firms such that we can better explain the effects coming from the variance in ESG scores.

Our general model to find this relationship can be illustrated as:

$$y_{i,t} = \alpha_i + \beta_1 ESG_{i,t} + \theta X'_{i,t} + u_{i,t}, \quad u_{i,t} \sim IID(0, \sigma_u^2) \quad (2.3)$$

where $y_{i,t}$ denotes our volatility measure or idiosyncratic risk measure, respectively. α_i is the unknown intercept for each firm (fixed effect), $ESG_{i,t}$ is our measure of ESG score and

explanatory variable, thus, making β_1 our main coefficient of interest. $X_{i,t}$ is a vector of control variables and $u_{i,t}$ is the time-varying idiosyncratic error term (Wooldridge, 2018).

As we also want to dig deeper into what specific pillars of the ESG scores that drives the various effects we can extend our model to include measures of the environmental-, social- and governance scores. We are able to use all the three pillars as explanatory variables as the pillars are orthogonal meaning they are mutually exclusive subcategories of the total ESG score (Bannier et al., 2019).

Thus, our new extended model can be illustrated as:

$$y_{i,t} = \alpha_i + \beta_1 E_{i,t} + \beta_2 S_{i,t} + \beta_3 G_{i,t} + \theta X'_{i,t} + u_{i,t} \quad u_{i,t} \sim IID(0, \sigma_u^2) \quad (2.4)$$

where the variables are the same as in Equation 2.3 except for $E_{i,t}$, $S_{i,t}$ and $G_{i,t}$ now being our explanatory variables meaning that β_1 , β_2 and β_3 are our coefficients of interest.

When analysing panel data the individual-specific effect, denoted as α_i , is allowed to be correlated with the explanatory variables. This correlation is acceptable because the α_i can be effectively eliminated by differentiating it away as it does not vary over time.

The most important condition for our model is that the error term, $u_{i,t}$, must not be correlated with the explanatory variables in any of the periods, t . Additionally, for the model to hold, the explanatory variables must undergo changes over time, and these changes should not occur with identical values from one period to another. So in a fixed effect model any explanatory variable that is constant over time for all i gets swept away. This means we would not be able to include variables such as industry or location of the firm in the regression model. Therefore no such explanatory variables are included in our model. In addition to the classical assumptions for OLS estimation, panel data analysis necessitates the assumption of homoscedasticity, meaning that the variance of the error term is constant across all observations in the panel (Wooldridge, 2018). Heteroscedasticity can be controlled for by estimating the model with robust standard errors. In our empirical analysis 4.2.2 we will further explain why we use cluster robust standard errors, which takes into account heteroscedasticity. When the assumptions are met and α_i is gone we can run the regression without worrying about biased results.

To sum up, when we use a fixed effect transformation, we analyze variations in intercepts among individuals while assuming consistent slopes and constant variance across groups. When using fixed effects, we don't need an assumption on α_i . It is allowed to be correlated with other regressors, since an individual specific effect is time-invariant and considered a part of the intercept (Park, 2011). However, if it is the case that α_i is not correlated with the explanatory variables, and one still transforms this away by using fixed effects, then the results will be inefficient (Wooldridge, 2018). This last argument leads to a discussion whether or not the fixed effect is the most appropriate estimation to use.

2.4.1 Fixed Effect vs Random Effect

When discussing a fixed effect estimation, it is inevitable not to address the concept of a random effect model as well. Even though the two models are very similar, there are important differences to take into account when figuring out which estimator to use.

In a random effect estimation, it is assumed that the individual effect (α_i) is uncorrelated with any regressor (where the opposite applies for fixed effects). For a random effect model, the intercept and slopes of the regressors remain consistent across individuals. The variation among individuals or time periods resides in their individual-specific errors rather than in their intercepts (Park, 2011). An advantage of using random effects is that we can include explanatory variables that are constant over time, since random effects assume that the unobserved effect α_i is not correlated with some of the explanatory variables (Wooldridge, 2018; Park, 2011)

In our case, we estimate the fixed effects model, because we believe it is best suited for our analysis and 2.3 and 2.4 will give unbiased fixed effects estimates if $u_{i,t}$ is not correlated with the explanatory variables in all included time periods. Oikonomou et al., 2012 also use a fixed effect estimator and argue that it is the most suitable specification for analysing the S&P500. Also, to support their argument, they test their model with the Hausman Specification test, thus, we will also perform this test. The Hausman specification test compares a random effect model to the fixed effect model. The null-hypothesis is that individual effects α_i are uncorrelated with any regressor in the model. So if the null hypothesis is not rejected, a random effect model is favored over its fixed counterpart (Park, 2011). We run the Hausmann

test in STATA and find a p-value < 0.05 for all our regressions.² This means that we reject the null-hypothesis and a fixed effect estimation is most suitable.

3 Data

Our dataset consists of numerable variables from the 503 stocks offered by 500 large U.S. companies in the S&P500 Index, and covers the period January 2003 to January 2023.

We extract all ESG measures of the companies from Refinitiv Eikon as it lets us extract all historical ESG ratings of the companies such that we are able to account for changes in ESG scores within our regressions. All other variables for the S&P500 stocks are extracted from Bloomberg. Lastly, for our Fama-French 5-Factor regression we extract the necessary factors from Kenneth French's webpage updated in cooperation with Dartmouth College (French, 2023).

The reason why we chose this sample period is that ESG scores for the S&P500 firms are not available on Refinitiv Eikon before 2003. Not all firms have scores each period, which is why we drop observations, when nothing is reported. This means we allow an unbalanced panel at the expense of possible methodological and computational complication (Park, 2011). To get balanced data we could entirely drop those periods where some companies have nothing reported, however, as STATA is able to handle unbalanced panels, we find it much more important to not throw any usable information away.

3.1 Portfolio Set-Up

In order to find the effect of greater ESG scores on the average excess return by estimating the Fama-French 5-Factor model, we divide our data into 5 portfolios dependent on their respective ESG scores. Thus, we divide our sample into quintiles where the first quintile Q1 have the lowest ESG ratings and the last quintile Q5 have the highest ESG ratings. We divide our sample into these five quintiles for each month such that companies that increase or decrease their ESG scores get reallocated to the corresponding quintile for the respective month. This ensures that the Q5-portfolio always consists of the companies with the highest ESG ratings and the Q1-portfolio always consists of the companies with the lowest ESG

²Full test statistics can be found in Appendix [A5](#), [A6](#), [A7](#) & [A8](#)

ratings. Thus, we are able to make sure that changes in the ESG scores does not affect or distort our regression results.

Furthermore, in accordance with (Bannier et al., 2019) and (Carhart, 1997) we create a sixth Long-Short portfolio. This portfolio is long in the high ESG-rated Q5-portfolio and short in the low ESG-rated Q1-portfolio. Thus, the Long-Short portfolio will receive the return of the Q5-portfolio whereas it will lose the return of the Q1-portfolio, leaving the overall return of the Long-short portfolio to be the difference between the Q5-portfolio and the Q1-portfolio. With this portfolio we are able to capture the difference in the abnormal return of the high and low rated quintile-portfolios where a positive α suggests greater return from high ESG-rated companies whereas a negative α suggests a greater return from low ESG-rated companies.

3.2 Descriptive Statistics

In deciding which variables to include in regressions and how to proxy them, we follow other studies, especially (Bannier et al., 2019; Cai et al., 2016; Oikonomou et al., 2012 and Monti et al., 2022).

Summary statistics for the variables used in our regressions are presented in table 1. We have sorted the statistics into the quintiles we also used for the Fama-French regression, Q1 being the portfolio with the worst rated ESG scores and Q5 being the one with the best. The mean, standard deviation and number of observations for each variable are also presented.

Table 1: Descriptive Statistics

	Q1	Q2	Q3	Q4	Q5	Mean	N	SD
ESG score	26.270	40.799	50.966	61.351	75.120	50.848	96705	20.37
Combined ESG score	25.891	39.690	48.609	55.928	64.487	46.878	96705	18.63
Environmental score	11.948	27.640	42.486	57.054	72.136	42.187	96705	29.49
Social score	29.804	42.300	52.293	63.153	77.082	52.875	96705	22.50
Governance score	34.851	50.527	56.967	62.981	74.439	55.909	96705	21.61
Controversies Score	91.191	87.661	83.837	74.713	65.744	80.655	96693	30.82
Volatility	2.060	1.984	1.933	1.852	1.770	1.919	95558	1.02
Idiosyncratic Risk	1.667	1.553	1.505	1.440	1.376	1.508	95558	0.79
Size	9.006	9.472	9.814	10.250	10.644	9.836	94947	1.45
Leverage	0.591	0.618	0.636	0.674	0.672	0.638	68248	0.25
Profit	0.028	0.024	0.025	0.024	0.024	0.025	94908	0.03
Dividend Indicated Yield	1.812	2.290	2.465	2.487	2.823	2.421	75096	1.73
Return	0.015	0.013	0.012	0.012	0.011	0.013	95459	0.09

Source: Refinitiv Eikon, Bloomberg and own calculations

Note: All scores regarding ESG and controversies measures are represented in percentages

Other than extracting our dependent variables (volatility and idiosyncratic risk) and independent variables (different ESG-scores), we extract a range of control variables. The variables of volatility and idiosyncratic risk are directly available in Bloomberg. Historical volatility, as explained in 2.2.2, refers to the measurement of market volatility based on the annualized standard deviation of relative price changes. It is computed by considering the daily closing prices over the preceding calendar year. Idiosyncratic risk refers to the unsystematic risk calculated as the volatility of the stock return which is not explained by the company's β as determined by CAPM.

To measure firm size, we take the natural logarithm of total assets, the leverage variable is derived by dividing liabilities by total assets, while profitability is calculated by dividing operating income by total assets. Finally, the Dividend Indicated Yield is sourced directly from Bloomberg.

The average ESG score in Q1 is 26.3 pct. and for Q5 it is 75.1 pct. The overall mean of the ESG score is 50.848. It is worth noticing that the environmental score overall is a bit lower than the social and governance scores. Naturally, we see that the environmental, social and governance scores increase with each quintile. However, what might seem a little

more contradictory is that the controversies score decreases with each quintile. A possible explanation for the decreasing controversies score might be that companies that are subject to a greater risk of being part of a scandal may allocate more resources to score high in all of other the three pillars to avoid being picked out as a controversial company or to mitigate a bad reputation from previous scandals. Another explanation could be that companies with high ESG scores just simply get affected more strongly by controversies due to the principle "The higher you fly, the harder you fall" as hypothesised by (Dorfleitner et al., 2020). As the controversies score decreases with the quintiles we therefore see a greater deviation between the ESG score and the Combined ESG score as we move up the quintiles.

The descriptive statistics gives us the first indications of what the relationship between ESG scores, return and risk is. We see that the return decreases for each quintile from the worst to the best rated and that both the volatility and idiosyncratic risk fall as well. Thus, the summary statistics overall support our hypothesis that higher ESG scores provide lower returns but also lower risk.

We see that there exists a positive correlation between a firm's size and its ESG rating performance. Moreover, the ESG rating tends to have a positive association with both the dividend indicated yield and leverage, whereas profitability exhibits a slight decline.

4 Empirical Analysis and Results

In our empirical analysis we estimate the Fama-French 5-Factor model and our fixed effect model on risk and go into depth with the results. For the Fama-French we estimate the model for each portfolio created as explained in 3.1.

4.1 Fama-French 5-Factor Estimation

When estimating the Fama-French 5-Factor model for each of the quintiles sorted by ESG score we find the following results shown in Table 2.

Table 2: Fama-French 5-Factor by ESG Score

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
Mkt_RF	1.0113*** (0.0279)	1.0265*** (0.0244)	1.0284*** (0.0224)	1.0334*** (0.0221)	0.9718*** (0.0186)	-0.0395 (0.0237)
SMB	0.2157*** (0.0503)	0.2343*** (0.0490)	0.1319* (0.0548)	0.0576 (0.0402)	-0.0629* (0.0300)	-0.2786*** (0.0488)
HML	0.0439 (0.0439)	0.1325*** (0.0371)	0.2419*** (0.0366)	0.1666*** (0.0311)	0.1609*** (0.0369)	0.1170** (0.0358)
RMW	0.1172* (0.0557)	0.1099 (0.0593)	0.0321 (0.0568)	0.0769 (0.0464)	0.0749 (0.0451)	-0.0423 (0.0483)
CMA	-0.2048** (0.0636)	-0.0403 (0.0523)	-0.0176 (0.0528)	0.0314 (0.0555)	0.1147* (0.0545)	0.3195*** (0.0534)
α	0.0057*** (0.0009)	0.0040*** (0.0007)	0.0031*** (0.0008)	0.0025*** (0.0007)	0.0020*** (0.0006)	-0.0036*** (0.0009)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Looking at the parameter for the abnormal return, α , we see that we get a positive and highly significant estimate for all our quintiles close to what was estimated in (Bannier et al., 2019). By investing in each of the quintile-portfolios our model predicts that you would be able to get a positive abnormal return for all of them. For the Q1-portfolio an α -estimate of 0.0056 is equivalent to 0.56% meaning that investing in the Q1-portfolio on average will yield a monthly return approximately 0.56%-points higher than the Fama-French 5-Factor model predicted. This is a rather high additional percentage to receive on a monthly basis.

However, if we compare the α 's to the ones found by a standard CAPM-regression as can be seen in Appendix A4 we find on average smaller α 's which indicates that the model better predicts the excess return and the remaining α will to a larger degree only capture the effect of the portfolio selection through ESG scores as explained in Section 2.3.

In addition to relatively high ESG scores, only having positive α 's suggests that all the quintile-portfolios are underrated. The reason for these very high and only positive α 's is most likely that the excess market return also known as the risk premium in our model is for all firms incorporated in the U.S. As the return for the stocks in the S&P500 index is higher,

there is a greater part of the return that is not predicted by the rest of the variables, thus, leading to higher α 's. Even though all the α 's are positive for the quintile-portfolios we also see that there is a rather stable trend of decreasing abnormal returns when moving up in quintiles.

Thus, the 5-Factor model suggests that the higher ESG rating the lower the return. This is also suggested by our Long-Short portfolio as we see that betting on the portfolio with high ESG rating and against the portfolio with low ESG ratings yields a negative abnormal return. Similar to Bannier et al., 2019 we also find that the negative excess return found in the Long-Short portfolio is not driven by a negative return from investing in firms with high ESG scores, but rather by an extremely high positive return from investing in firms with low ESG scores.

Since the ESG consists of three different components it would be interesting to see if all the components have the same trend as the overall score or if there is any deviations from the individual pillars. We therefore run the same regression but with the quintiles sorted by the environmental, social and governance score, respectively. The α 's of each regression can be seen in Table 3 and the full regression results can be found in Appendix: A1, A2 & A3.

Table 3: Fama-French 5-Factor, α by Environmental, Social & Governance

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
α_E	0.0069*** (0.0009)	0.0037*** (0.0009)	0.0017* (0.0008)	0.0026** (0.0009)	0.0018** (0.0006)	-0.0052*** (0.0010)
α_S	0.0049*** (0.0009)	0.0040*** (0.0007)	0.0028*** (0.0008)	0.0032*** (0.0007)	0.0023*** (0.0006)	-0.0027** (0.0009)
α_G	0.0043*** (0.0008)	0.0035*** (0.0007)	0.0038*** (0.0007)	0.0031*** (0.0007)	0.0025*** (0.0007)	-0.0018 (0.0010)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

When we run the regression with our quintiles sorted by the environmental score we find a somewhat similar decreasing trend of α . What is noticeable about this regression is that the Q3-portfolio has the lowest α followed by quite a high α in the Q4-portfolio in comparison and then quite a low α once again in the Q5-portfolio. This gives some indication that it might not be a strictly negative correlation between environmental score and return. However, once

again we should take into consideration that it is difficult to assess the changes in abnormal return solely by the α of the different quintiles. We therefore look at our LS-portfolio as it can speak to the significance of the difference between abnormal returns of the high and low ESG portfolios. Compared to Table 2 we see a much larger α for our Q1-portfolio and a slightly lower α for our Q5-portfolio. We therefore get a much more negative abnormal return from our LS-portfolio which indicates that the effect of having a high environmental score has a greater negative effect on the return.

When we run the regression with our portfolios sorted after their social score instead we similarly find a decreasing trend of α . Once again we find that the Q4-portfolio is slightly higher than the Q3-portfolio and the Q5-portfolio although it is less pronounced. Compared to Table 2 we find an α a bit lower for the Q1-portfolio and an α a bit higher for the Q5-portfolio. Oppositely than the environmental regression we therefore get a less negative abnormal return from the LS-portfolio indicating that the effect of having a high social score has a less negative effect on the return.

Lastly, when we run our regression with our portfolios sorted after their governance score we once again find a decreasing trend of α . However, in this regression the α of the Q1-portfolio is much lower and the α of the Q5-portfolio is much higher and we therefore get a much less negative abnormal return of our LS-portfolio. Furthermore, when we look at the LS-portfolio we see that in this case it is not significant indicating that the effect of having a high governance score does not have a significant negative effect on the return.

Overall, we therefore find that the abnormal return decreases for all the three pillars, although it was not significant for the governance pillar. The environmental had the biggest decrease in α and is therefore the driving factor of the decreasing α in the ESG score.

From these three single pillar regressions we see that the result from the ESG regression is affected differently by the different pillars. It would therefore be interesting to estimate an additional regression that includes the controversies score to see how this affects the results

Table 4: Fama-French 5-Factor by Combined ESG Score

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
Mkt_RF	1.0309*** (0.0295)	1.0132*** (0.0222)	0.9970*** (0.0199)	1.0336*** (0.0211)	0.9965*** (0.0210)	-0.0344 (0.0228)
SMB	0.1902*** (0.0560)	0.1821*** (0.0484)	0.1327** (0.0413)	0.0742 (0.0411)	-0.0024 (0.0381)	-0.1926*** (0.0452)
HML	0.0553 (0.0487)	0.1766*** (0.0379)	0.2053*** (0.0311)	0.2177*** (0.0337)	0.0911* (0.0423)	0.0359 (0.0395)
RMW	0.1207 (0.0626)	0.0224 (0.0540)	0.0622 (0.0491)	0.1038* (0.0480)	0.1016 (0.0517)	-0.0191 (0.0499)
CMA	-0.1608* (0.0675)	0.0103 (0.0489)	-0.0343 (0.0509)	-0.0520 (0.0542)	0.1201* (0.0561)	0.2809*** (0.0506)
α	0.0056*** (0.0009)	0.0039*** (0.0007)	0.0032*** (0.0007)	0.0023** (0.0007)	0.0023*** (0.0007)	-0.0033*** (0.0009)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

In Table 4 we see the regression results when we sort our portfolios after the ESGC score that includes the controversies score. The α in this regression does not differ substantially from those found in Table 2 but we do see slightly lower α 's on average across Q1-Q4 and a slightly larger increase in the α of Q5. Primarily due to the increase of α in Q5 we therefore see a slightly less negative abnormal return of the LS-portfolio. As we saw from Table 1 the controversies score decreased in the quintiles making the Combined ESG score deviate the most from the ESG score in Q5. Thus, it was to be expected that there would only be small deviations in abnormal return in the lower quintiles and the largest deviation in Q5. This estimation might yield slightly more accurate results since it takes the increased controversies into account. In conclusion an abnormal return of -0.33% will be our best guess of the effect of ESG score on return.

As described in Section 2.2.1 the return of a specific stock or portfolio is largely dependent on the underlying risk of said asset. As α theoretically only consists of the abnormal return coming from ESG it should likewise be a proxy of the required abnormal return given the risk only coming from ESG. Thus, it might be that the decreasing trend of α is attributable to a proportionate decrease in our risk measures (Bannier et al., 2019). This leads us to further

examine the correlation between ESG scores and risk measures.

4.2 Fixed Effect Estimation

We continue our analysis by doing a fixed effect estimation of the model from Equation 2.3. We follow (Bannier et al., 2019; Dunn et al., 2017; Jo and Na, 2012; Monti et al., 2022) by regressing the ESG score on our different measures of risk.

We run the fixed effect regression in STATA with clustered robust standard errors at the firm level, because we could be worried that there would be correlated shocks within the firms. By estimating clustered robust standard errors, our analysis acknowledges the presence of correlation within clusters and adjusts our standard errors accordingly. Also, because we are dealing with panel data, there might be autocorrelation present, which means that the errors are not independent and exhibit some form of systematic correlation. This is what we allow for when estimating with cluster robust standard errors. These errors will be higher than the classic standard errors and thereby provide us with more accurate estimates of p-values and confidence intervals. Thereby, our statistical tests and inferences become more reliable and robust against potential bias arising from clustering effects Wooldridge, 2001. The results from our estimation can be seen in Table 5.

4.2.1 Control variables

As presented in the descriptive analysis, we include control variables in our fixed effect estimation. In order to find out which control variables were most important to include, we follow (Bannier et al., 2019 Cai et al., 2016 Oikonomou et al., 2012 and Monti et al., 2022) and their arguments.

Control variables are important for our analysis to more accurately estimate the effect of sustainability on risk of firms. It helps us to isolate the relationship between our variables of interest by holding effects of other variables constant and thereby making it possible for us to treat the data as random.

We include firm size (measured as log of total assets) as a control variable when estimating the relationship between ESG score and risk of firms. This is commonly used and is important because larger firms tend to have more resources and greater capacity to manage their risks

compared to smaller firms. For example, a larger firm may have a higher ESG score and lower risk simply because it has more resources to invest in ESG practices and risk management. As expected, our regression analysis shows that firm size has a significant negative effect on volatility as can be seen in Table 5.

Profitability (which we find by dividing operating income with total assets) is also a relevant control variable to include, as more profitable firms likewise have more resources and greater capacity to manage their risks. As expected we find that profit has a highly significant negative effect on both volatility and idiosyncratic risk.

We also include leverage (total liabilities divided by total assets) since when a company's leverage increases, the likelihood of it being unable to meet its debt obligations increases as well. As expected we find a highly significant and positive relationship.

Lastly, we add the dividend indicated yield as a control variable, which has a positive and significant effect on both volatility and idiosyncratic risk. This however, is not as straightforward intuitively, because often it is ambiguous whether it has a negative or positive effect on risk (Blume, 1980; Oikonomou et al., 2012). On one hand, a high dividend yield may be seen as an indicator of financial stability, since you might view a company as stable if it consistently pays out a significant portion of its earnings as dividends. On the other hand, a high dividend yield can also be an indicator of financial distress or limited growth opportunities.

4.2.2 Effect of ESG on Risk

Since we found the Combined ESG score to be more accurate we continue using this in our estimations. When we do the regression of Combined ESG score on volatility we find a significant negative correlation as can be seen in Table 5. When regressing on volatility, the coefficient estimate of $\hat{\beta}_1^{Vol} = -0.284$ suggests that for every one-unit increase in the Combined ESG score, there is a 0.284 unit decrease in volatility, all else being equal. As our ESG scores are percentages that run from 0 to 1 the decrease in volatility is equivalent to $\frac{0.284}{100} = 0.00284$ for a 1-point increase in ESG score. Therefore, a one standard deviation increase in Combined ESG score reduces stock volatility by $0.00284 * 18.63 = 0.0529$ and

by dividing this with the mean of the volatility³ we get that it results in a $\frac{0.0529}{1.919} = 2.76\%$ decrease in volatility. These findings imply that companies with higher ESG scores are likely to experience lower volatility compared to those with lower ESG scores.

Table 5: Combined ESG score on Risk

	Volatility	Idiosyncratic Risk
Combined ESG score	-0.284* (0.132)	-0.034 (0.102)
Leverage	0.526** (0.180)	0.423** (0.155)
Size	-0.125* (0.049)	-0.027 (0.042)
Profit	-5.239*** (0.678)	-3.597*** (0.496)
Dividend Indicated Yield	0.056** (0.021)	0.034* (0.017)
Constant	2.922*** (0.450)	1.439*** (0.380)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

When regressing the Combined ESG score on idiosyncratic risk, we do not find significant results, hence we cannot conclude, that there is the same negative relationship between ESG and idiosyncratic risk. Bannier et al., 2019 find similar results, which increases the robustness of our findings. It is important to note that these results are not entirely aligned with our initial hypothesis, which suggested that higher ESG scores would be associated with lower equity risk, both in terms of volatility and idiosyncratic risk.

Monti et al., 2022 evaluates how the systematic and unsystematic components of equity risk will be affected by CSR activities. They point out that idiosyncratic risk measures are most likely driven by operational risk drivers while systematic risk partly will mirror investor preferences and that it therefore makes sense to address the idiosyncratic risk measure alone. They find that both risk measures will be similarly affected by CSR activities which is contradicting to our findings as we find no effect on idiosyncratic risk. However, they also find that the risk-reducing effect on idiosyncratic risk of good ESG practices is about 13

³See Table 1

times stronger in civil-law countries (such as many European countries) compared to common law countries (such as the U.S., England, and Australia). Thus, we would might be able to find a significant negative effect on idiosyncratic risk if we used a European sample, and our contradicting results may be due to a geographical bias.

4.2.3 Effect of Environmental, Social & Governance on Risk

Once again, since the ESG score consists of three different components it is interesting to investigate which specific factors that are the primary drivers for the observed decrease in volatility. Furthermore, we are interested in exploring whether any of these factors have a significant impact on idiosyncratic risk as well. We therefore change the regression model to that of Equation 2.4 where we regress environmental, social and governance score on volatility and idiosyncratic risk, respectively. The results can be seen in Table 6.

Table 6: Environmental, Social & Governance on Risk

	Volatility	Idiosyncratic Risk
Environmental	-0.746*** (0.152)	-0.472*** (0.112)
Social	0.689*** (0.178)	0.624*** (0.140)
Governance	-0.239* (0.120)	-0.134 (0.097)
Leverage	0.539** (0.169)	0.424** (0.142)
Size	-0.118* (0.051)	-0.034 (0.045)
Profit	-5.278*** (0.669)	-3.654*** (0.494)
Dividend Indicated Yield	0.054** (0.021)	0.032 (0.017)
Constant	2.822*** (0.458)	1.452*** (0.390)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Based on the results, it is evident that there is a highly significant negative relationship between the environmental score and both volatility. The coefficient estimate of $\hat{\beta}_1^{Vol} = -0.746$

suggests that for every one-unit increase in environmental score, volatility will decrease by $\frac{0.746}{100} = 0.00746$, all else being equal. Compared to the estimate found for an increase in Combined ESG score this is quite a negative effect. In comparison a one-standard deviation increase in environmental score is equivalent to a $\frac{29.49 \times 0.00746}{1.919} = 11.46\%$ decrease volatility. We also find a significantly negative estimate for the idiosyncratic risk: $\hat{\beta}_1^{Idio} = -0.472$. Similarly, this is also quite a negative effect compared to the effect of an increase in Combined ESG score that had no significant effect on idiosyncratic risk. A one-standard deviation increase in environmental score is equivalent to a $\frac{29.49 \times 0.00472}{1.508} = 9.23\%$ decrease in idiosyncratic risk.

This suggests that investing in companies with a high environmental score reduces both the volatility and idiosyncratic risk. Since we in the Fama-French regression in Section 4.1 found that a high environmental score had the most negative effect on return it is fitting that it also has a significant negative effect on risk as a compensation.

Regarding the influence of the social score, we get highly significant and positive estimates, indicating that firms with a higher social score will have higher volatility and idiosyncratic risk. The estimates are $\hat{\beta}_2^{Vol} = 0.689$ and $\hat{\beta}_2^{Idio} = 0.624$ equivalent to a $\frac{22.5 \times 0.00689}{1.919} = 8.1\%$ and $\frac{22.5 \times 0.00624}{1.508} = 9.3\%$ increase in the risk measure for a one standard deviation increase in the social score. This is in line with the findings from (Bannier et al., 2019), where it is clear that for U.S. companies environmental concerns appear to be much more important than social. The result seems to vary across different countries as the paper finds a negative relationship between the social score and equity risk, when using a European sample. A negative effect from the social pillar on equity risk in Europe is also found by (Sassen et al., 2016). Hence, it is important to distinguish companies from different geographic places, which was also argued by (Monti et al., 2022). It is possible that European companies have made more progress in incorporating factors from the social component into their decisions than U.S. companies. This would lead to the U.S. companies having higher risk for higher social scores and the other way around for European companies.

Furthermore, investor preferences may differ for U.S. and European companies. If investors in U.S. companies place less emphasis on social considerations in their investment decisions, it might contribute to the positive relationship between social score and risk. Monti et al., 2022 argue that systematic risk will partly mirror investor preferences, which could be one possible

explanation that the social component does not have a negative effect on risk in the U.S., but that other papers have found that relationship for the European market. Comparing our findings to the results from the Fama-French regression in Section 4.1 we find that the social score has a negative effect on return and positive effect on risk. This indicates that the low return from investing in companies with high social scores is not compensated by lower risk, but rather a disproportionately higher risk.

Regarding the governance score, we find a significant estimate of $\hat{\beta}_3^{vol} = -0.239$ on the volatility measure, which is what we would expect since better management should entail more stable results which should lower equity risk. On the other hand, we get insignificant results when regressing on idiosyncratic risk. It is somewhat surprising there is not a perfectly clear relationship between good governance practices and how effective it is in reducing risk. However, our results are in accordance with Bannier et al., 2019. The reason for this might be, as noted by both (Nguyen, 2011; Lubatkin and O’neill, 1987), that risk-taking in strategic management plays a big role in growing the company. Furthermore, as firms grow they gain more market power which in turn makes them less exposed to volatility in the market, thus, increasing their idiosyncratic risk.

Comparing our findings to the results from the Fama-French regression in Section 4.1 we find that the governance score has an insignificant effect on return, a slightly negative effect on volatility, and an insignificant effect on idiosyncratic risk. These results suggest that investing in companies with high governance score could be beneficial since it does not have a substantial impact on returns, but can contribute to slightly lower risk.

4.3 Risk-Adjusted Returns

Having examined the impact of ESG on both return and risk individually, we now explore the relationship between these two measures by constructing a return-to-risk ratio. This ratio allows us to assess the efficiency of ESG investments in generating returns relative to the associated level of risk. By quantifying the trade-off between return and risk, we gain further insights into the risk-adjusted performance of ESG portfolios.

By dividing our α ’s of each quintile with the risk measure of the corresponding quintile we can estimate a return-to-risk ratio to see if the lower abnormal returns of high Combined

ESG-portfolios are adequately compensated through lower risk.

Table 7: Return-to-Risk Ratios by Combined ESG Score

	Q1	Q2	Q3	Q4	Q5
$\frac{\alpha}{\text{Volatility}}$	0.271	0.194	0.167	0.123	0.129
$\frac{\alpha}{\text{Idiosyncratic Risk}}$	0.335	0.246	0.214	0.158	0.166

Note: The α in this table has been multiplied by 100 to express the percentage-point increase in abnormal return of a 1-unit increase in the risk measure.

As we see from Table 7 there is a declining trend of return-to-risk ratios. Even though the portfolios with higher ESG scores have lower risk it is not enough to account for the much higher abnormal return from the portfolios with lower ESG scores. For a 1-unit increase in volatility and idiosyncratic risk the Q1-portfolio has the highest increase in abnormal return of 0.271 and 0.335 respectively. Thus, the low ESG portfolios more than compensates for the higher risk attributed with an investment which indicates that it is more efficient to invest in the low ESG portfolios.

Table 8: Return-to-Risk Ratios by Individual ESG Pillars

		Q1	Q2	Q3	Q4	Q5
Environmental	$\frac{\alpha}{\text{Volatility}}$	0.340	0.177	0.088	0.144	0.100
	$\frac{\alpha}{\text{Idiosyncratic Risk}}$	0.419	0.229	0.111	0.187	0.130
Social	$\frac{\alpha}{\text{Volatility}}$	0.247	0.205	0.143	0.169	0.126
	$\frac{\alpha}{\text{Idiosyncratic Risk}}$	0.306	0.260	0.182	0.219	0.162
Governance	$\frac{\alpha}{\text{Volatility}}$	0.210	0.180	0.199	0.166	0.137
	$\frac{\alpha}{\text{Idiosyncratic Risk}}$	0.264	0.229	0.253	0.212	0.175

Note: The α in this table has been multiplied by 100 to express the percentage-point increase in abnormal return of a 1-unit increase in the risk measure.

Table 8 shows the same return-to-risk ratios for the α and associated risk for each of the three ESG pillars individually. Similarly, we find a decreasing return-to-risk ratio for all the ESG

pillars which further suggests that low ESG portfolios are more efficient.

Both return-to-risk ratios for environmental decrease quite substantially, especially from Q1 to Q2. Thus, we see that even though the environmental score had the most negative effect on risk, the effect of larger returns from low environmental scores are much greater.

From social we see that both return-to-risk ratios decrease quite a bit as well. This was very much expected as we through our analysis found higher social scores to both decrease return and increase risk. However, as we saw in Table 3 the decrease in α is not as large for social as compared to environmental. Thus, we see a greater effect on the return-to-risk ratios from the decreasing α 's compared to the change in risk.

Lastly, we see a small decrease in return-to-risk ratios from governance. This was also expected as we saw only a small decrease in returns from governance and similarly a small decrease in risk that was only significant for volatility. Thus, an increase in governance score neither affects return nor risk much, which is why we find only a small decrease in the return-to-risk ratios.

Once again, we find environmental to have the greatest effect of the three pillars. Thus, the environmental pillar is the primary driver of the decreasing return-to-risk ratio of ESG suggesting that investing in portfolios with low environmental scores will be the most efficient.

5 Discussion

While our analysis and findings align with many existing studies on the subject, there are still several aspects that lead to further discussion and exploration.

5.1 Limitations of our data

Other risk measures could be important for understanding the impact of ESG score on risk. Many studies (Bannier et al., 2019; Monti et al., 2022; Boffo and R. Patalano, 2020; Oikonomou et al., 2012; Cai et al., 2016) also analyze the firms' downside risk. Downside risk can be measured in multiple ways and many use the variables Value-at-risk (VaR) and conditional value-at-risk (CVaR). VaR and CVaR are risk measures that specifically address

the downside risk associated with stock prices, capturing more extreme negative movements, where CVaR provides a measure of the average magnitude of losses beyond the VaR level (Wipplinger, 2007). VaR and CVaR might have been a good addition to our understanding of the impact of sustainability practices on risk. Since we have not included these variables in our analysis, we leave out some relevant insights to the relationship between ESG activity and risk of firms. If we had used VaR and CVaR we would gain a better understanding of the potential losses that investors face. Unlike standard deviation, which captures both the upside and downside of returns, VaR and CVaR specifically focus on the downside risk, providing a more comprehensive evaluation of potential losses in stock investments.

Another thing that could affect our results is the time frame of our data. It is mostly in recent years that ESG has emerged and become a larger consideration for investors. As described in Section 3 our data starts in 2003 as this was the first point in time where there was a substantial reporting and measuring of ESG. However, the concept of ESG was still fairly new for both companies and investors. It was first in 2014 that SRI (Socially Responsible Investing) became a well-known concept with approximately 30% of all assets under management now being under SRI, which was an increase of more than 60% from 2012 (Eccles et al., 2016). Thus, the years up until 2014 might not have been that indicative of the true value of SRI as some companies did not see the need of engaging in ESG reporting which can have distorted our results. Furthermore, as mentioned by (Drei et al., 2019) the SRI in 2014 mainly consisted of the environmental and governance pillar, whereas the turning point for the social pillar came even later around 2016. It can therefore be argued that the data after that period yields a more accurate relationship between ESG, return and risk. However, we believe that the data would be too limited if we only included recent years. The relationship between ESG activities and return and risk should be further analysed in the future when additional data is available.

Even though we find that the risk-adjusted return is decreasing for each of the quintiles (going from low ESG scoring to high), we will not conclude that there cannot be any advantages of investing in companies with high ESG scores. Although our results indicate that ESG is associated with lower risk-adjusted return, it can also function as a form of insurance for systematic risks. Monti et al., 2022 address the importance of CSR having a risk-reducing effect and that this relationship is increasing in riskier periods. To support this, they divide

their sample into three different periods: Before the financial crisis (2002-2006), during (2007-2009) and after the financial crisis (2010-2015). They find that in 2002-2006, CSR seemed to have no effect on risk, in 2007-2009 some risk measures were influenced and most interestingly, that after the financial crisis, all their risk measures were strongly influenced by CSR. They emphasize that investing sustainably will give investors a kind of insurance, as it will protect against risk of rare and large shocks such as the financial crisis and the COVID-19 period. So even though ESG comes with the cost of lower risk-adjusted returns, ESG investments will still be valuable in turbulent times because of the provided insurance. This suggests limitations from our analysis, since we do not cover the aspect of different time periods and especially risk measures such as VaR and CVaR, which captures effects of more extreme movements.

The measure of ESG ratings is another aspect of limitations of our data that could have influenced our results. As there are many different providers of ESG scores and they have different methods of calculating these ratings, the choice of the ESG measure could change the results (Dorfleitner et al., 2015). Similarly, the choice of country for our sample could have had a large impact on our findings. As mentioned in Section 4.2.3 the results between U.S. and European samples differ quite substantially. Thus, running our regression of indices from other countries could yield different findings. Additionally, running the regression on the S&P1500 instead might also alter our results as the inclusion of more companies can affect our different variables.

5.2 Method

We have chosen to estimate our model using fixed effects, which we believe provides a suitable approach for handling our data. However, it is worth noting that alternative estimation methods could have been considered. One such method involves incorporating a lagged dependent variable into the model. This approach is often employed by researchers working with panel data. It is possible to use both a fixed effect estimator and lagged dependent variables at the same time, but this requires conditioning on both the lagged dependent variable and α_i . Using both at the same time is much more demanding for the conditions for consistent estimation, which is why we have chosen to keep our estimation simple by only using fixed effects (Angrist and Pischke, 2009).

Also, we argue that our fixed effect estimator is unbiased, since we believe that the idiosyncratic error term $u_{i,t}$ is not correlated with any of the explanatory variables in any of the periods, t . It is difficult to say that this assumption will be completely fulfilled, but we believe that we have taken into account most of the possible complications that could arise with the assumption. For example, we have chosen a wide range of control variables which we argued were some of the most important ones to consider. If these are chosen correctly, it will reduce possible biases in our estimates (Wooldridge, 2001). However, it is possible that we have left out important control variables, which could have influenced our estimates.

In our analysis, a possible extension to consider is the influence of industries when making investment decisions related to ESG. The significance of industries cannot be overlooked, as different industries may yield different effects. While it is commonly observed that ESG factors have a negative impact on equity risk, there have been hypotheses suggesting the opposite. Cai et al., 2016 develops a "resource-constraint" hypothesis about how some firms in controversial industries, such as alcohol, tobacco, and gambling, may not utilize Corporate Environmental Responsibility (CER) as a long-term strategy nor as a tool to mitigate negative effects and unfavorable public perception. Any attempts to increase their environmental impact through CER activities may backfire as the public perceive it as a wasteful use of resources. In accordance to this, (Jo and Na, 2012) propose the "window-dressing" hypothesis. Firms in controversial industries might make a distrustful attempt to increase their stock price or improve their reputation through CSR activities. If the investors find out their true intentions, the CSR engagements may end up increasing firm risk. Thus, if we had looked at effects across industries, we might have found that ESG engagement would have affected return and risk differently.

6 Conclusion

In this paper we have estimated a Fama-French 5-Factor regression (Fama and French, 2014) for different quintile-portfolios sorted by ESG scores following (Bannier et al., 2019). Our results indicate that a portfolio consisting of stocks with the lowest Combined ESG scores results in an abnormal return of 0.56%, while a portfolio with the highest Combined ESG scores results in an abnormal return of 0.23%. This yields a significant negative abnormal return of -0.33% in our Long-Short portfolio. This significant difference in abnormal returns

suggests a negative impact of investing in stocks with higher ESG scores. Running the same regression with the quintiles sorted after environmental-, social-, and governance scores, respectively, we find that the environmental score was the leading factor in the negative relation between ESG score and abnormal return.

Due to the relationship between return and risk we further examined the relationship between ESG and risk by regressing the Combined ESG score on both overall volatility and idiosyncratic risk following among others (Bannier et al., 2019; Oikonomou et al., 2012). We find a negative effect of the ESG score on volatility, aligning with previous research findings. This prompted us to investigate further to identify the specific factors within ESG that contribute to the lower risk. Splitting the ESG score into its individual pillars, we observed that the environmental score had a highly negative and significant effect on both risk measures. The governance score showed a lower, but still negative effect on volatility. In contrast, the social score exhibited a highly positive and significant effect on both risk measures. Once again, we find that environmental pillar is the driving factor of the negative relationship between ESG and risk across U.S. companies.

Lastly, we compare the benefit of the higher return to the higher associated risk of investing in lower rated ESG portfolios. We find a decreasing trend of return-to-risk ratios across both our risk measures, thus, the return of the low rated ESG portfolios more than compensates for the increased risk. Similarly, we find that all of the three individual pillars have decreasing return-to-risk ratios with the environmental score having the largest decrease. Overall, we therefore find it to be more efficient to invest in companies with lower ESG ratings and that of the three pillars it is most crucial to consider the firm's environmental score. However, as noted in Section 5.1 companies with high ESG ratings might still be a viable investment option as it provides more insurance in turbulent times.

7 Appendix

Table A1: Fama-French 5-Factor by Environmental

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
Mkt_RF	0.9812*** (0.0282)	1.0414*** (0.0293)	1.0305*** (0.0234)	1.0320*** (0.0258)	0.9768*** (0.0171)	-0.0044 (0.0274)
SMB	0.2180*** (0.0572)	0.2398*** (0.0543)	0.1624*** (0.0447)	-0.0242 (0.0490)	-0.0354 (0.0326)	-0.2533*** (0.0537)
HML	0.0131 (0.0461)	0.2375*** (0.0414)	0.1349*** (0.0316)	0.1512*** (0.0387)	0.2578*** (0.0329)	0.2447*** (0.0433)
RMW	0.0589 (0.0629)	0.0588 (0.0691)	0.2051** (0.0621)	0.0257 (0.0570)	0.1099* (0.0440)	0.0510 (0.0563)
CMA	-0.1895** (0.0678)	-0.0612 (0.0651)	-0.0389 (0.0513)	0.0445 (0.0545)	0.0845 (0.0560)	0.2741*** (0.0645)
α	0.0069*** (0.0009)	0.0037*** (0.0009)	0.0017* (0.0008)	0.0026** (0.0009)	0.0018** (0.0006)	-0.0052*** (0.0010)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A2: Fama-French 5-Factor by Social

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
Mkt_RF	1.0070*** (0.0253)	1.0358*** (0.0244)	1.0249*** (0.0236)	1.0196*** (0.0188)	0.9841*** (0.0217)	-0.0229 (0.0231)
SMB	0.2068*** (0.0462)	0.1668*** (0.0427)	0.1780** (0.0553)	0.0543 (0.0450)	-0.0287 (0.0330)	-0.2355*** (0.0455)
HML	0.0803 (0.0521)	0.0812* (0.0343)	0.2064*** (0.0356)	0.2728*** (0.0361)	0.1050** (0.0318)	0.0247 (0.0420)
RMW	0.1297* (0.0550)	0.0913 (0.0576)	0.1469* (0.0585)	0.0290 (0.0511)	0.0144 (0.0462)	-0.1154* (0.0479)
CMA	-0.1211 (0.0653)	-0.0161 (0.0551)	-0.0788 (0.0544)	-0.0474 (0.0531)	0.1461** (0.0513)	0.2672*** (0.0620)
α	0.0049*** (0.0009)	0.0040*** (0.0007)	0.0028*** (0.0008)	0.0032*** (0.0007)	0.0023*** (0.0006)	-0.0027** (0.0009)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A3: Fama-French 5-Factor Governance

	Q1	Q2	Q3	Q4	Q5	LS-Portfolio
Mkt_RF	1.0574*** (0.0283)	1.0061*** (0.0210)	1.0165*** (0.0206)	0.9889*** (0.0211)	1.0025*** (0.0207)	-0.0548* (0.0263)
SMB	0.1798* (0.0784)	0.1546*** (0.0382)	0.1384** (0.0423)	0.0919** (0.0336)	0.0128 (0.0364)	-0.1669* (0.0644)
HML	0.0633 (0.0411)	0.1385*** (0.0366)	0.1329*** (0.0345)	0.1850*** (0.0426)	0.2267*** (0.0383)	0.1634*** (0.0468)
RMW	0.0435 (0.0740)	0.0475 (0.0485)	0.0872 (0.0514)	0.1044* (0.0495)	0.1297** (0.0493)	0.0861 (0.0590)
CMA	-0.0956 (0.0651)	-0.0922 (0.0519)	0.0517 (0.0544)	-0.0126 (0.0546)	0.0308 (0.0585)	0.1264 (0.0653)
α	0.0043*** (0.0008)	0.0035*** (0.0007)	0.0038*** (0.0007)	0.0031*** (0.0007)	0.0025*** (0.0007)	-0.0018 (0.0010)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A4: CAPM Regression by ESG Score

	Q1	Q2	Q3	Q4	Q5
Mkt_RF	1.0600*** (0.0313)	1.0842*** (0.0327)	1.0830*** (0.0354)	1.0565*** (0.0266)	0.9646*** (0.0235)
α	0.0057*** (0.0009)	0.0042*** (0.0009)	0.0030** (0.0010)	0.0027** (0.0008)	0.0024** (0.0008)

Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A5: Hausman (1978) specification test (Volatility, Equation [2.3](#))

	Coef.
χ^2 -value	66.7
p-value	0

Table A6: Hausman (1978) specification test (Idiosyncratic Risk, Equation [2.3](#))

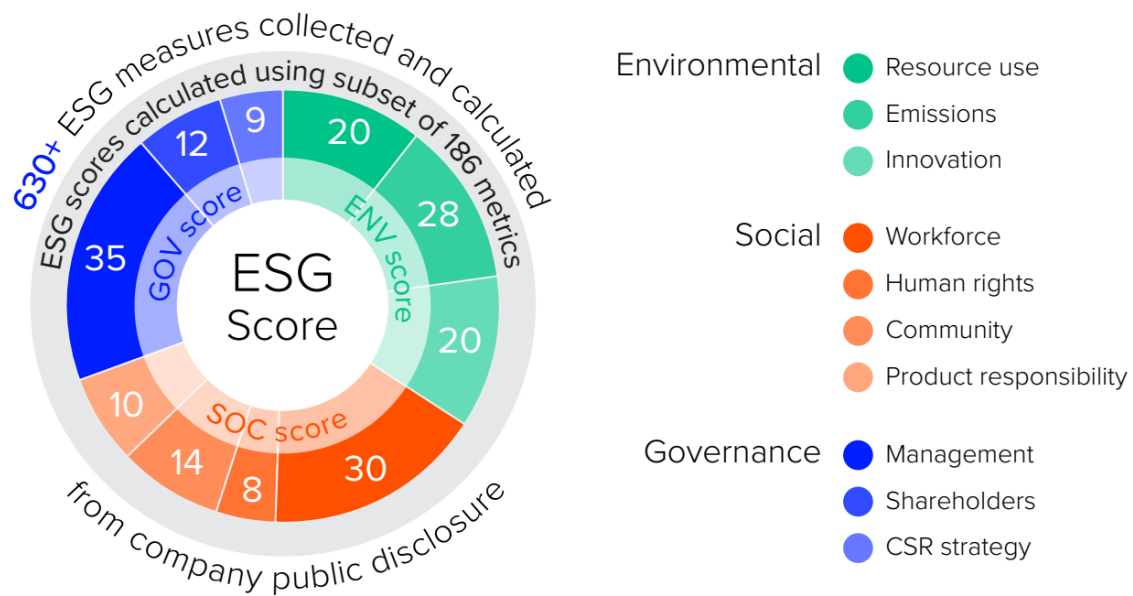
	Coef.
χ^2 -value	42.76
p-value	0

Table A7: Hausman (1978) specification test (Volatility, Equation [2.4](#))

	Coef.
χ^2 -value	109.86
p-value	0

Table A8: Hausman (1978) specification test (Idiosyncratic Risk, Equation [2.4](#))

	Coef.
χ^2 -value	67.16
p-value	0

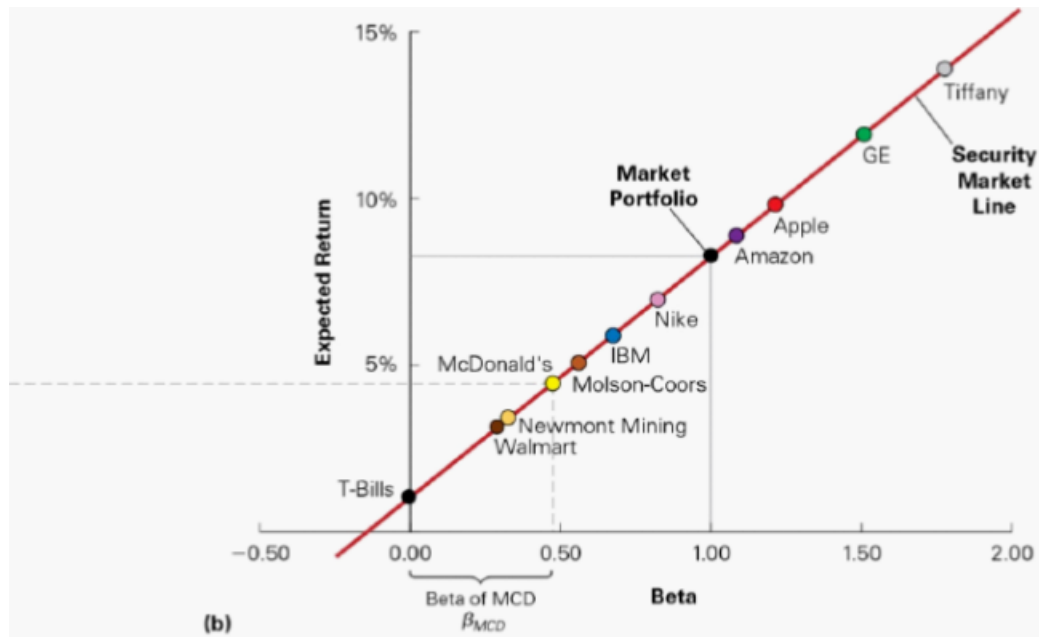
Figure A1: ESG Score Structure

(a) Source: Refinitiv, 2022

Figure A2: ESG Score Materiality Matrix

Pillars	Catagories	Themes
Environmental	Emmission	Emissions
		Waste
		Biodiversity*
		Environmental management systems*
	Innovation	Product innovation
		Green revenues, research and development (R&D) and capital expenditures (CapEx)
	Resource use	Water
		Energy
		Sustainable packaging*
		Environmental supply chain*
Social	Community	Equally important to all industry groups, hence a median weight of five is assigned to all
	Human rights	Human rights
	Product responsibility	Responsible marketing
		Product quality
		Data privacy
	Workforce	Diversity and inclusion
		Career development and training
		Working conditions
		Health and safety
Governance	CSR strategy	CSR strategy
		ESG reporting and transparency
	Management	Structure (independence, diversity, committees)
		Compensation
	Shareholders	Shareholder rights
		Takeover defenses

(a) Source: Refinitiv, 2022

Figure A3: Security Market Line

(a) Source: Berk and DeMarzo, 2020

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